

CY 511P: Physical Chemistry Laboratory

Course Code	CY 511P
Course Name	Physical Chemistry Laboratory
L-T-P-C	0-0-8-4
Prerequisites	Undergraduate level Physical Chemistry Laboratory
Intended for	M.Sc
Distribution	Core
Semester	Odd/Even

Preamble

This course intends to teach students the experiments on kinetics, catalysis, spectroscopy, photochemistry and determination of few physical properties of molecules. This course also gives an opportunity to the students to gain expertise on different instrumental techniques.

Course Modules

• Module I:

1. Calibration of volumetric apparatus.
Spectroscopy
2. Analysis of the vibrational spectra of HCl, CCl₄, small organic molecules.
3. Simultaneous Determination of chromium and manganese in a mixture by visible light spectroscopy
Physical Property and Surface chemistry
4. Determination of critical micellar concentration.
5. Determination of pK_a of an amino acid by pHmeter
Distribution Law
6. Distribution coefficient of I₂ between two immiscible solvents.
7. Determination of the equilibrium constant of the reaction $KI + I_2 = KI_3$ using the result from the previous experiment.

• Module II:

Electro-analytical Method

8. The potentiometric titration of an acid mixture
9. Conductometric titrations of strong acid HCl using strong base NaOH
10. Conductometric titrations of weak acid CH₃COOH using strong base NaOH
11. Conductometric titration of a triple mixture of HCl, NH₄Cl and KCl by NaOH and AgNO₃.

• Module III:

Phase Equilibria

12. Determination of phase diagram of a simple eutectic system (Naphthalene – Biphenyl, Naphthalene-Diphenyl amine)
13. Determination of phase diagram of a binary solid system forming a compound (e.g., Naphthalene – m-dinitrobenzene)

Photochemistry

14. Interaction of protein- fluorescence of protein complex, Kinetic study of protein activity of *p*-nitro phenol acetate to *p*-nitrophenol conversion, Fluorescence spectrum and stern-volmer quenching constant.

Nanomaterials

15. Synthesis of gold nanoparticles and characterization by UV-VIS, DLS techniques etc
16. Size Effect of Gold Nanoparticles in Catalytic Reduction of *p*-Nitrophenol with NaBH₄

References

1. *Experimental physical chemistry*, F. A. Bettelheim
2. *Experimental physical chemistry*, G. P. Matthews
3. *Practical physical chemistry*, by Alex. Findlay.
4. *Experimental Physical Chemistry*, D. P. Shoemaker, C. W. Garland, and J. W. Nibler
5. B. Viswanathan & R.S. Raghavan, *Practical Physical Chemistry*, Viva Books, 2009.
6. A. Ghosal, B. Mahapatra, A. K. Nad, *An Advanced Course in Practical Chemistry*, New Central Book Agency Pvt Ltd, Calcutta (2000).